Automatic Speech Recognition: A Brief History and Recent Trends

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Abstract—Automatic speech recognition (ASR) has evolved through several technical eras over seven decades: template matching, statistical HMM/GMM modeling, hybrid DNN-HMM acoustic models, fully end-to-end neural architectures, and today's self-supervised and weakly supervised foundation models. Since 2023, ASR has increasingly fused with large language models into unified, real-time multimodal systems. This report traces that history and surveys these recent trends.

Index Terms—automatic speech recognition, hidden Markov models, deep neural networks, end-to-end speech recognition, self-supervised learning, foundation models

I. Introduction

Automatic speech recognition (ASR), the conversion of spoken audio into text, underlies dictation, voice assistants, captioning, and speech translation. Its dominant paradigm has shifted repeatedly, from template matching to statistical, hybrid, end-to-end, and foundation-model approaches. Section II traces this history (Table I); Section III surveys current trends.

II. Historical Development

The earliest ASR systems recognized small, speaker-dependent vocabularies via template matching. Bell Labs' Audrey [1] used vowel formants to recognize digits at ∼97–99% accuracy for one talker, and Itakura later combined linear predictive coding with dynamic time warping for isolated-word recognition (1975). ARPA's Speech Understanding Research program (1971–1976) targeted connected speech: CMU's Harpy exceeded its 90%-accuracy goal via finite-state beam search, while Hearsay-II introduced the "blackboard" architecture.

From the 1980s, ASR converged on HMMs with GMM emissions, trained via Baum-Welch/Viterbi; IBM's Tangora and CMU's Sphinx demonstrated large-vocabulary and speaker-independent recognition. Shared corpora (TIMIT, Wall Street Journal, Switchboard) enabled benchmarking, while discriminative training (MMI, MPE) pushed Switchboard WER from ∼43% (1995) to 15–25% by the mid-2000s, GMM-HMM's practical ceiling.

TABLE I
Eras of Automatic Speech Recognition

Era	Years	Key Method / System
Template matching	1950s–76	Formant/DTW; Audrey, Harpy
Statistical GMM-HMM	1980s–2000s	HMM+GMM, n-gram LM; Sphinx, HTK
Hybrid DNN-HMM	2009–13	CD-DNN-HMM; Bing, Google Voice
End-to-end neural	2014–20	CTC, RNN-T, Conformer
Foundation era	2019–26	Self-sup. pretraining, audio LLMs; Whisper, GPT-4o

III. Recent Trends

Neural acoustic models re-entered ASR when DNN-HMM systems outperformed GMM-HMM baselines, cutting Switchboard WER from 27.4% to 18.5% and enabling industrial deployment by 2012. Connectionist Temporal Classification then enabled training directly on unsegmented audio-label pairs; attention-based seq2seq models, the RNN-Transducer, and the Conformer followed, replacing separately trained acoustic, pronunciation, and language models with one end-to-end network.

Since 2019, self-supervised pretraining on unlabeled audio (wav2vec 2.0, HuBERT, WavLM) has cut labeled-data needs, while Whisper [2] (680,000 hours of weak multilingual supervision) generalizes zero-shot across noise and accents. Massively multilingual and distilled variants now extend coverage and enable on-device streaming. Most recently, audio-native LLMs (GPT-4o, Gemini, Moshi) fuse recognition directly with generation, removing cascaded ASR-LLM-TTS latency, while parallel work targets atypical-speech robustness and semantic evaluation beyond raw WER.

IV. Conclusion

ASR has progressed from template matching through statistical, hybrid, and end-to-end architectures to today's self-supervised, multimodal foundation models, and is now converging toward multilingual, streaming, LLM-integrated spoken-language understanding.

References

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